Def. Let G be a group. Define

$$G^{(0)} = G \quad \text{and} \quad G^{(i+1)} = [G^{(i)}, G^{(i)}] \quad (i \in \mathbb{N}).$$

The descending chain

$$G = G^{(0)} \trianglerighteq G^{(1)} \trianglerighteq G^{(2)} \trianglerighteq \cdots$$

is called the derived series of G.

Thm. Let G be a group. Then,

G is solvable if and only if $G^{(r)} = 1$ for some $r \in \mathbb{N} \cup \{0\}$.

Proof. Suppose that G is solvable. That is, there is a subnormal series

$1 = H_r \triangleleft H_{r-1} \triangleleft \cdots \triangleleft H_1 \triangleleft H_0 = G$ such that H_i / H_{i+1} is abelian for all $0 \le i \le r-1$.

Since H_i / H_{i+1} is abelian if and only if $H_i' = [H_i, H_i] \le H_{i+1}$, so

G/H_1 is abelian $\Leftrightarrow$ $G' = G^{(1)} \le H_1$

H_1/H_2 is abelian $\Leftrightarrow$